

Mahesh Chitakoti

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SUMMARY

Backend & AI Engineer with **2.5+ years of experience** building high-throughput microservices, scalable APIs, and production GenAI applications. Specialized in **Python, FastAPI, RAG architectures, vector databases, and cloud infrastructure**. Reduced LLM inference costs by **35%**, cut median API latency by **60% (800ms to 320ms)**, and engineered backend automation systems processing **1K+ daily workflows**.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI & GenAI: RAG, Agentic Workflows, Semantic Search, Embeddings, Prompt Engineering, LangChain, OpenAI API, Google Gemini

AI Infrastructure: Qdrant, pgvector, Vector Databases

Backend: FastAPI, Node.js, Express.js, NestJS, REST APIs, Microservices, WebSockets, MQTT

Databases: PostgreSQL, Redis, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD

Frontend: React.js, Next.js, Redux, Tailwind CSS

EXPERIENCE

AI Engineer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

Remote

- Architected and deployed production **Retrieval-Augmented Generation (RAG)** and LLM-powered features using **FastAPI, LangChain, Gemini APIs, and Qdrant**, reducing LLM inference costs by **35%**.
- Developed AI microservices for transcript processing, semantic search, and document retrieval, leveraging vector similarity search to support production LLM and knowledge-retrieval workflows.
- Engineered resilient real-time streaming workflows using **WebSockets** and **MQTT**, implementing retry and fallback mechanisms to improve reliability across production messaging systems.
- Built AI analytics dashboards using **React.js** and **Redux**, improving system observability and enabling faster production incident diagnosis.

Full Stack Developer – The DevOps Team

Oct 2023 – Aug 2025

Bangalore, India

- Optimized high-traffic **Node.js** and **Express.js** APIs using Redis caching and database query optimization, reducing median response time from **800ms to 320ms**.
- Developed job recommendation workflows matching candidate skills, preferences, and job requirements to improve personalized role discovery.
- Built **React.js** and **Next.js** onboarding experiences with A/B testing to improve user activation and signup conversion.
- Implemented real-time notification systems using **WebSockets** and supported **Docker- and Kubernetes-based** production deployments in collaboration with DevOps teams.

PROJECTS

WorkBridge – AI-Powered Collaboration Platform — Personal Project

May 2026 – Present

React.js, Next.js, Node.js, Express.js, Python, PostgreSQL, Redis, OAuth | Live Project

- Designed and developed an AI-powered collaboration platform connecting students with industry projects through skill-based matching and real-time collaboration.
- Built backend APIs using **Node.js, Express.js, PostgreSQL, and Redis** for authentication, project management, collaboration, and messaging.
- Integrated AI recommendation services for project matching using **RAG, semantic search, and agentic workflows** to improve project-matching relevance.

CERTIFICATIONS

Oracle Agentic AI Certified Foundations Associate – Oracle

Aug 2026

Oracle AI Vector Search Certified Professional – Oracle

Sep 2025

Machine Learning I – Columbia University

Aug 2025

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional – Oracle

Aug 2024

EDUCATION

Adichunchanagiri Institute of Technology

Chikkamagaluru, KA

Bachelor of Engineering, Visvesvaraya Technological University (VTU)

Aug 2023